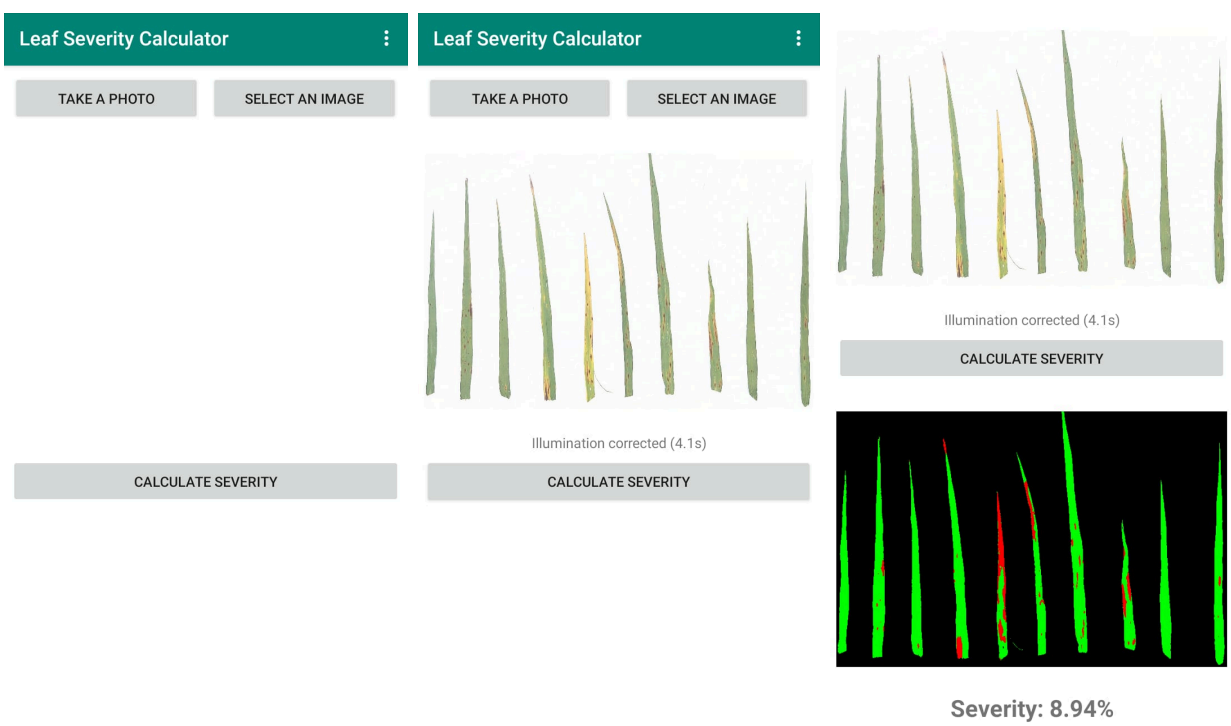
Leaf Severity Calculator - OpenCV-based Branch

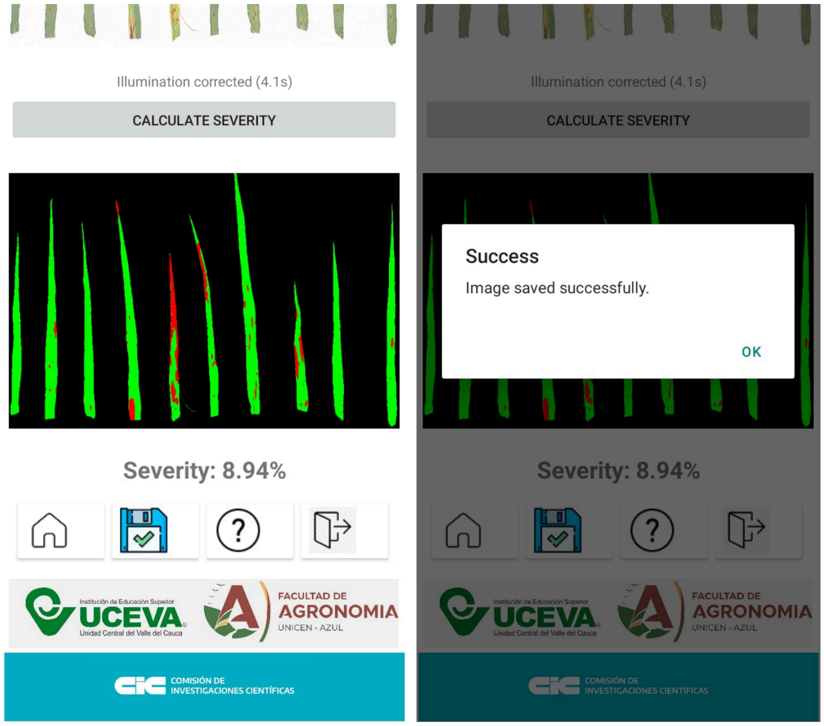
Version: 0.0.9b1 | **Stack**: Python 3.8 + Briefcase 0.3.19 | **Status**: ✅ Legacy Stable

Overview

Cross-platform Android application for analyzing barley leaf disease severity. This **OpenCV-based branch** maintains a **Python 3.8 legacy stack** with pre-compiled wheels for broader compatibility with older Android devices.

App Screenshots





Key Features

- 📷 **Capture & Upload**: Device camera or gallery image selection
- 🔬 **Threshold-Based Segmentation**: Uses fixed thresholds on blue channel (`ub`) and index `(green-red)/(green+red)` (`ui`); these values were calibrated offline using Otsu/K-means during model setup
- 📊 **Severity Calculation**: Real-time disease percentage computation
- 💾 **Save Results**: Export processed images to `/sdcard/Download/LeafSeverityImages`
- 🌐 **Full English USA UI**: All strings localized to English
- 📱 **Broad Device Support**: Python 3.8 compatible with older Android devices
- 🎯 **Multi-ABI Support**: arm64-v8a, armeabi-v7a, x86_64

Why This Branch?

Aspect	Main (0.0.9a1)	OpenCV-based (0.0.9b1)
Python Version	3.12 (Latest)	3.8 (Legacy)
Briefcase	0.3.22 (Current)	0.3.19 (Stable)
Dependencies	PyPI (Latest)	Pre-built wheels (cp38)

Aspect	Main (0.0.9a1)	OpenCV-based (0.0.9b1)
Device Compatibility	Modern devices (API 24+)	**Broader range**
Performance	Optimized	Stable, reliable
Use Case	New apps, modern devices	Legacy devices, stability

Choose this branch if: You need to support older Android devices or require the proven stability of Python 3.8 + OpenCV 4.5.1 + NumPy 1.19.5 stack.

Branch Processing Notes (Main vs OpenCV-based)

Both branches keep the same user-facing workflow:

- Camera/gallery image selection
- Illumination correction before severity calculation
- Aspect-ratio-preserving resize (fit inside 800x600)
- Threshold segmentation and severity display

Implementation differs by branch:

- **main**
    - Illumination correction: `numpy-rolling-ball`
    - Processing style: NumPy-first pipeline
    - Android stack: newer API/memory requirements (cp312 ecosystem)

- **OpenCV-based**
    - Illumination correction: `opencv-rolling-ball` (`cv2_rolling_ball`)
    - Processing style: OpenCV for decode/resize/encode + NumPy masks for severity math
    - Android stack: legacy compatibility (cp38 wheels, older device support)

This split is intentional and reflects compatibility/memory constraints between modern and legacy Android targets.

Technology Stack

Component	Version	Source	Note
Python	3.8	Local venv (.venv38)	Python 3.12 not compatible with cp38 wheels
Briefcase	0.3.19	Local venv	Matches Briefcase template v0.3.19
toga-android	~0.4.5	PyPI	
NumPy	1.19.5	Local wheels (cp38)	Pre-built for Android ABIs
opencv-python	4.5.1.48	Local wheels (cp38)	Pre-built for Android ABIs
opencv-rolling-ball	Latest	PyPI	
tatogalib	0.9.6	Local wheel (universal)	File browser integration

Project Structure

```
leafseveritycalculator/
├── .venv38/                    # Python 3.8 virtual environment (loca
├── src/leafseveritycalculator/
│   ├── __init__.py
│   ├── app.py                  # Main application logic
│   ├── __main__.py
│   └── resources/              # App resources
├── android/
│   └── app_template/
│       └── src/main/
│           ├── AndroidManifest.xml     # Permissions config
│           └── res/            # Drawables, layouts, strings
├── tests/
│   ├── __init__.py
│   ├── _support.py
```

```
│   ├── test_app.py
│   └── leafseveritycalculator.py
├── wheels/                          # Local pre-built Android wheels
│   ├── numpy-1.19.5-0-cp38-cp38-android_21_arm64_v8a.whl
│   ├── numpy-1.19.5-0-cp38-cp38-android_16_armeabi_v7a.whl
│   ├── numpy-1.19.5-0-cp38-cp38-android_21_x86_64.whl
│   ├── opencv_python-4.5.1.48-1-cp38-cp38-android_21_arm64_v8a.whl
│   ├── opencv_python-4.5.1.48-1-cp38-cp38-android_16_armeabi_v7a.whl
│   ├── opencv_python-4.5.1.48-1-cp38-cp38-android_21_x86_64.whl
│   ├── tatogalib-0.9.6-py3-none-any.whl
│   └── numpy_rolling_ball-1.0.0-py3-none-any.whl (universal)
├── pyproject.toml                   # Project configuration + ABI-organize
└── CHANGELOG
```

Why there is a second README/LICENSE inside `leafseveritycalculator/`

That folder is the **Briefcase app project root** (packaging root for Android builds). Keeping `README.rst` and `LICENSE` there is normal for BeeWare/Briefcase projects, while root-level docs are for repository-level information.

- Root docs/files: repository documentation for contributors and GitHub
- `leafseveritycalculator/README.rst` and `leafseveritycalculator/LICENSE`: app packaging metadata and project-local docs

The duplication is acceptable and expected in this repo layout.

Getting Started

Prerequisites

⚠️ **IMPORTANT**: This branch requires **Python 3.8**, not 3.12.

- **Python 3.8.x** (Windows, macOS, or Linux)
  - Download: python.org/downloads/release/python-3810/

- **Android SDK** (API 21+)
- **Android NDK**

- **Java Development Kit (JDK)** 11+
- **Briefcase 0.3.19** (not 0.3.22)

Installation & Setup

1. **Clone the repository**:

```
git clone https://github.com/alencina-faa/Leaf-Severity-Calculator_androi
cd Leaf-Severity-Calculator_android
git checkout OpenCV-based  # Switch to this branch
```

2. **Create Python 3.8 virtual environment**:

```
# Windows
python3.8 -m venv .venv38
.venv38\Scripts\activate

# macOS/Linux
python3.8 -m venv .venv38
source .venv38/bin/activate
```

3. **Install dependencies**:

```
pip install --upgrade pip setuptools wheel
pip install briefcase==0.3.19
```

⚠️ **Do NOT upgrade Briefcase** - version 0.3.19 is required for this stack.

Building for Android

Development/Debug Build

```
cd leafseveritycalculator

# Ensure .venv38 is activated
# On Windows: .\.venv38\Scripts\activate
```

```
# On macOS/Linux: source ../.venv38/bin/activate

# Initial setup
.venv38\Scripts\python -m briefcase create android --no-input

# Build debug APK
.venv38\Scripts\python -m briefcase build android

# Package debug APK
.venv38\Scripts\python -m briefcase package android
```

Output: `dist/LeafSeverityCalculator-0.0.9b1.apk`

Release Build

```
cd leafseveritycalculator

# Update dependencies (rebuilds with local wheels)
.venv38\Scripts\python -m briefcase update android -r

# Package release AAB (for Google Play)
.venv38\Scripts\python -m briefcase package android
```

Output: `dist/LeafSeverityCalculator-0.0.9b1.aab` (67.15 MB with all ABI wheels)

ABI Support

This branch includes local wheels for **three Android ABIs**:

ABI	API Level	Architecture	Primary Device	Wheel Status
arm64-v8a	21	64-bit ARM	Modern Android phones	✅ Included
armeabi-v7a	16	32-bit ARM	Older Android phones	✅ Included
x86_64	21	64-bit x86	Emulators, tablets	✅ Included

Build output: APK/AAB contains Python + wheels for **all ABIs** (~67 MB). Chaquopy automatically selects the correct wheel for each device's architecture.

Local Wheels Management

Why Local Wheels?

NumPy 1.19.5 and OpenCV 4.5.1.48 are old packages with broken setuptools build backends (missing `setuptools.extern.six`). Pre-built `.whl` files bypass the build process entirely.

Wheel Organization (by ABI)

In `pyproject.toml`, wheels are organized explicitly:

```
[tool.briefcase.app.leafseveritycalculator.android]
requires = [
    "toga-android~=0.4.5",
    # NumPy & OpenCV wheels - organized by ABI for explicit matching
    # ARM64-v8a (API 21)
    "../wheels/numpy-1.19.5-0-cp38-cp38-android_21_arm64_v8a.whl",
    "../wheels/opencv_python-4.5.1.48-1-cp38-cp38-android_21_arm64_v8a.whl",
    # ARMv7a (API 16)
    "../wheels/numpy-1.19.5-0-cp38-cp38-android_16_armeabi_v7a.whl",
    "../wheels/opencv_python-4.5.1.48-1-cp38-cp38-android_16_armeabi_v7a.whl",
    # x86_64 (API 21)
    "../wheels/numpy-1.19.5-0-cp38-cp38-android_21_x86_64.whl",
    "../wheels/opencv_python-4.5.1.48-1-cp38-cp38-android_21_x86_64.whl",
    # Universal wheels
    "../wheels/tatogalib-0.9.6-py3-none-any.whl",
    "opencv-rolling-ball",
]
```

Adding/Updating Wheels

1. Download wheel from Chaquopy PyPI index

2. Place in `wheels/` directory
3. Add reference in `pyproject.toml` under appropriate ABI section
4. Rebuild: `.venv38\Scripts\python -m briefcase update android -r`

Key Dependencies Explained

NumPy 1.19.5

- **Why 1.19.5**: Last version compatible with Python 3.8 + Briefcase 0.3.19
- **Pre-built wheels**: For cp38 across all ABIs (arm64-v8a, armeabi-v7a, x86_64)
- **Usage**: Array operations, channel extraction, mathematical processing

OpenCV 4.5.1.48

- **Why 4.5.1**: Stable release compatible with legacy Briefcase stack
- **Pre-built wheels**: For cp38 across all ABIs
- **Usage**: Image filtering, rolling-ball background subtraction (via opencv-rolling-ball)

Briefcase 0.3.19

- **Template version**: Matches `v0.3.19` from BeeWare GitHub
- **Do NOT upgrade**: Version 0.3.22 breaks Python 3.8 compatibility

Running on Device

1. **Activate venv**:

   ```
   .venv38\Scripts\activate  # or source .venv38/bin/activate
   ```

2. **Build and install**:

   ```
   cd leafseveritycalculator
   .venv38\Scripts\python -m briefcase package android
   adb install dist/LeafSeverityCalculator-0.0.9b1.apk
   ```

3. **Grant permissions** (on first launch):

    - Camera access
    - Storage read/write
4. **Launch** from device app drawer

Development Notes

Testing

```
cd leafseveritycalculator
.venv38\Scripts\python -m pytest tests/
```

Code Structure

Main Application (`src/leafseveritycalculator/app.py`):

- `LeafSeverityCalculator`: App class
- `startup()`: Initialize UI
- `take_photo()`: Camera integration
- `open_image()`: Gallery file picker
- `extract_background_color()`: Illumination correction (rolling-ball)
- `process_image()`: Image segmentation + severity calculation
- `save_image()`: Save results

Android Build Configuration

Set in `pyproject.toml` `build_gradle_extra_content`:

```
android {
    defaultConfig {
        ndk {
            abiFilters 'arm64-v8a'
        }
    }

    signingConfigs {
        release {
```

```
            keyAlias "upload-key"
            keyPassword "android"
            storePassword "android"
            storeFile file("F:\\Users\\Alberto\\.android\\upload-key-leafsever
        }
    }

    buildTypes {
        release {
            signingConfig signingConfigs.release
            minifyEnabled true
            shrinkResources true
        }
    }
}
```

Troubleshooting

Issue: "Python 3.12 not compatible with wheel cp38-cp38-android_21_arm64_v8a"

Solution: Ensure you're using Python 3.8:

```
python --version  # Should print 3.8.x
.venv38\Scripts\python --version  # Activate venv and verify
```

Issue: "ModuleNotFoundError: setuptools.extern.six"

Solution: This error happens when building old numpy/opencv from source. Use pre-built wheels only. Don't try to install via pip:

```
# ❌ Wrong - don't do this:
pip install numpy==1.19.5
```

```
# ✅ Correct - wheels already in repo:
# (Wheels are automatically included via pyproject.toml)
```

Issue: "Briefcase command not found"

Solution: Ensure venv is activated:

```
.venv38\Scripts\activate
pip install briefcase==0.3.19
```

Issue: Build fails with "gradle build" error

Solution: Clean build artifacts and retry:

```
cd leafseveritycalculator
if (Test-Path "build\leafseveritycalculator\android") {
    Remove-Item -Recurse -Force "build\leafseveritycalculator\android"
}
.venv38\Scripts\python -m briefcase create android --no-input
.venv38\Scripts\python -m briefcase package android
```

Environment Configuration

Environment Variables

```
ANDROID_HOME        # Path to Android SDK
ANDROID_SDK_ROOT    # Alternative Android SDK path
ANDROID_NDK_ROOT    # Path to Android NDK
JAVA_HOME           # Path to JDK 11+
```

.gitignore

Already configured to exclude:

- `.venv38/` - Local Python 3.8 environment

- `build/` - Build artifacts
- `dist/` - Build outputs
- `*.pyc` - Python bytecode

Version Information

- **Current**: 0.0.9b1 (Beta Release)
- **Branch**: OpenCV-based (Legacy stable stack)
- **Python**: 3.8.x
- **Briefcase**: 0.3.19
- **Last Updated**: April 24, 2026

Related Branches

- **main**: Modern Python 3.12 + Briefcase 0.3.22 stack
  - Latest dependencies from PyPI
  - Better performance on modern devices
  - See `main` branch README

Deployment to Google Play

1. **Ensure build succeeds**:

   ```
   .venv38\Scripts\python -m briefcase package android
   # Output: dist/LeafSeverityCalculator-0.0.9b1.aab (67.15 MB)
   ```

2. **Sign release** (already configured in `pyproject.toml`)

3. **Upload to Play Console**:

   - Navigate to Google Play Console
   - Select LeafSeverityCalculator
   - **Release** → **Testing** (recommended first) or **Production**
   - Upload `dist/LeafSeverityCalculator-0.0.9b1.aab`
   - Add release notes
   - Review and publish

Contributing

1. Create feature branch from `OpenCV-based`
2. Test with `.venv38` activated
3. Update version in `pyproject.toml` (PEP 440 format)
4. Update CHANGELOG if major changes
5. Commit and create pull request

License

See LICENSE file.

Authors

- Emiliano David
- Alberto Lencina
- Luisa Cabezas

Institution: Facultad de Agronomía, Universidad Nacional del Centro de la Provincia de Buenos Aires (UNICEN)

Email: alencina@azul.faa.unicen.edu.ar

Additional Resources

- BeeWare Briefcase v0.3.19 Docs
- Toga UI Toolkit
- Chaquopy Python-Android Build System
- NumPy 1.19.5 Release Notes
- OpenCV 4.5.1 Release Notes

Status: ✅ Stable Legacy | **Last Build**: LeafSeverityCalculator-0.0.9b1.aab (67.15 MB)